AMRITA VISHWA VIDYAPEETHAM

AHEAD Online

Statements in JDBC

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Objectives

- Statements in JDBC
- Resultset in JDBC

SQL Statements

Submitting an SQL statement

- Query compilation :checks the syntax
- Query plan generation and caching :select the best way to execute the query
- Query execution :execute the query in the best way that is selected previously.

JDBC Statements

- JDBC Statements are used to execute the SQL queries.
- The JDBC API provides us with three types of statements.
  - Statement interface
  - Prepared statement interface
  - Callable statement interface

Statement Interface

- It is used to implement simple statements with no parameters and return the ResultSet object.
- It is useful when we have static SQL queries which needs to be run at run time.
- No parameters can be sent in to retrieve the results based on the conditions.
- Statement stmt = conn.createStatement();

Prepared Statement Interface

- Using the parameterized statements.
- Pre-compiled SQL queries.
- The performance of the application will be faster if you use PreparedStatement interface because query is compiled only once.
- public PreparedStatement prepareStatement(String query)throws SQLException{}

Prepared Statement Interface

- String query = “select * from Students batch where batch.id =10”;
- PreparedStatement stmt = conn.prepareStatement(query);

Positional Parameters

- PreparedStatement stmt = conn.prepareStatement("INSERT INTO VALUES(?,?,?)");
- When we use PreparedStatement it contain query with positional parameters.
- Positional Parameters are specified using "?" symbol. Positional Parameters are used to supply the values to query.
- In-order to bind the values together with the parameters, we use setXXX() method.
- setInt(position, intvalue)

Callable Statement Interface

- Used to call stored procedures in a database.
- A stored procedure is like a function or method in a class, except it lives inside the database.

```
CallableStatement stmt =
    conn.prepareCall("{call calculateStatistics(?, ?)}");
  stmt.setInt (2, 123);
```

ResultSet

- The *java.sql.ResultSet* interface represents the result set of a database query.
- The SQL statements that read data from a database query, return the data in a result set

```
ResultSet rs = st.executeQuery("select * from account");
```

- When ResultSet object is created automatically a cursor will be created positioned before the first record.

```
public boolean next()
public xxx getxxx(int field_No)
public xxx getxxx(String field_Name)
```

Takeaways

- Statement in JDBC
- Resultset in JDBC

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Reference

[1]. Java How to Program Dietel,Harvey Tenth Edition

[2].https://docs.oracle.com/javase/tutorial/jdbc/basics